

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 521 — Past Questions & Materials
by Renance

Course: MAR 521
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 500

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Paper 1 of 1 — MAR 521 RUNNING AND MAINTENANCE OF SHIP POWER PLANT SECOND SEMESTER 2023/2024

Level 500 • Second Semester • PastQuestion

DEPARTMENT OF MARINE ENGINEERING

RUNNING AND MAINTENANCE OF SHIP POWER PLANT (MAR 521)

END OF SEMESTER TEST

TIME: 90 MINS. snk ee

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QUESTION ONE oo! eat

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4a. What constitutes "ship machinery spaces "? (3.5 marks)

LK». Discuss the role of main engines and auxiliaries onboard a ship (2.5 marks)

_7c. What do you understand by the term "Machinery space tools" ? (0.5 mark)

+d. Discuss two relevance of machinery spaces tools (1 mark)

,_ 8. Enumerate five basic tools for ship machinery's repair (2.5 marks)

QUESTION TWO

~ 4 Describe the investigation and remedial action required if a ship main engine turns on air but fails to fire on fuel? (2 marks)

YY b. With respect to machinery space knowledge, discuss the core functions of the engineering department onboard a ship (1 mark)

SY. How can we minimize maintenance tasks on ship machinery spaces? (i marks)

Vv What ensures high-functioning, clean and safe operations in ship's machinery spaces? (1 mark)

e. Discuss three possible ultimate yardsticks that can be used to measure the

V competence of a watchkeeping procedure that can improve the performance of a marine engineer (3 marks)

f. Having been taught "Running and maintenance of ship power plants" discuss the relevance of this knowledge on any Marine Systems of your choice (2 marks)

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by end users. A thermal plant with higher capacity is required to serve the community needs. Thus, this prompted DONGFANG INC. to design/manufacture a regenerative thermal plant (REG401) that would operate in RANKINE CYCLE.

As an EXPERT in marine steam turbine operations:

a) Explain the working principle of a regenerative cycle (1.5 mark)

Using the working diagram and T-S DIAGRAM below in Fig. 2:

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Superheater eae

my OM Turbine —i: W,, -

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mn | {1-m,)

Sax rm,

ii—m, -m,)

HP. heater (1 —m,)

{ 4m, . ; :

, “Fig.2: Regenerative cycle

i. ‘Neglecting pump work, provide the thermal efficiency equation for the

. regenerative cycle ; (3 marks)

ii. Assuming REG401 operate:as a SINGLE HEATER regenerative cycle where

-. steam enters the turbine at 30 bar and 400 degrees Celsius with exit pressure _

.of OAObar-and the feedwater heater is a direct contact type which’operate

‘at bar, deduce the efficiency of the thermal plant and increase in mean

- ‘\$émperature of heat-addition and show’by disparity the percentage rise due

+o regeneration (where the enthalpy of the steam at the point of entering

FWH is 2737.99KT/KG (8 marks)

QUESTION THREE

In a steam turbine, steam leaves the boiler plant at 20 bar, 300°C and expanded to 0.035 bar before entering the condenser, Compare the following cycles in terms of efficiencies and specific steam consumption for:

i. A-superheated Rankine cycles (2.5 marks)

ii, Areheat cycle with steam reheated to 300°C at the pressure when it becomes saturated vapour, (5 marks)